

Functions

$$y = 2x - 3 \quad \text{defined on } \mathbb{R} \text{ in } \mathbb{R}$$

$$y = x^2 \quad \text{defined on } \mathbb{R} \text{ in } [0, \infty)$$

$$y = \sqrt{x} \quad \text{defined on } [0, \infty)$$

$$y = 2x^3 - x^5 \quad \vdots \quad \text{in } [0, \infty)$$

$$y = \ln x$$

$$y = e^x$$

$$y = \sin x$$

$$y = \cos x$$

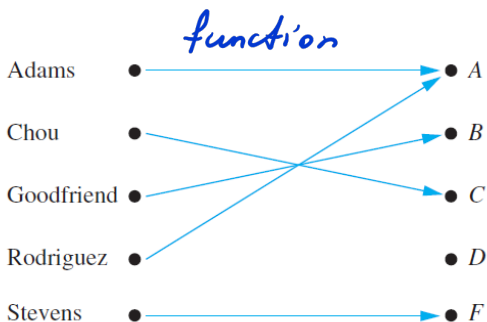
$$y = \tan x$$

$\vdots$

why?

$$\left. \begin{array}{l} y^2 = x \\ x^2 + y^2 = 1 \end{array} \right\} \text{curves}$$

In many instances we assign to each element of a set a particular element of a second set.



Let A and B be nonempty sets.

A **function** f from A to B (denoted as $f : A \rightarrow B$) is an assignment of exactly one element of B to each element of A .

We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

Element a is called **preimage** of b , and b is called an **image** of a .

- Functions are sometimes also called **mappings** or **transformations**.
- Set A is called **domain**.
- Set B is called **codomain**.
- Set $f(A) = \{b \in B \mid \exists a \in A \ f(a) = b\}$ is called **range or image** of the function $f : A \rightarrow B$.
- Image is a subset of codomain, that is $f(A) \subseteq B$.

Examples

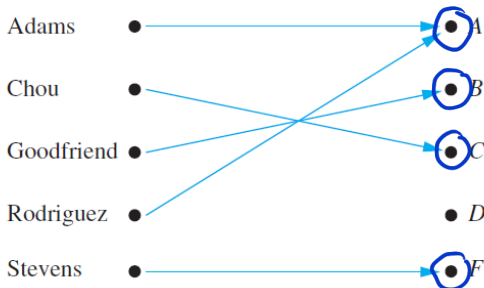


image of "Chou" is "C"

preimage of "F" is "Stevens"

preimage of "A" is {Adams, Rodriguez}

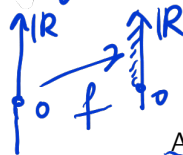
domain is {Adams, Chou, Goodfriend, Rodriguez, Stevens}

codomain is {A, B, C, D, F}

range is {A, B, C, F}

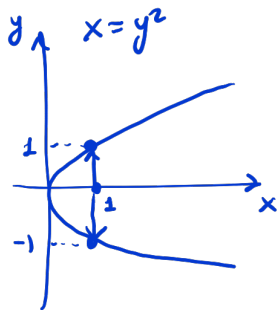
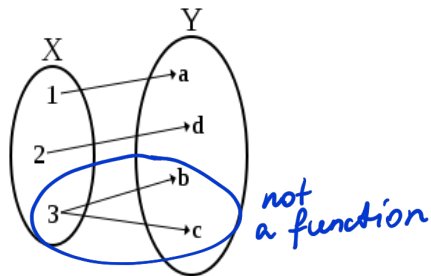
Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^2$.

Image is $[0, +\infty)$ and is a proper subset of codomain $\mathbb{R}$.



Answer Q1

Examples of not functions



Consider $y = f(x)$, where $x = y^2$ is not a function since $f(1) = 1$ and $f(1) = -1$.

Answer Q2

$f(x) = \sqrt{x}$ is not a function on $\mathbb{R}$ (from $\mathbb{R}$ to $\mathbb{R}$) since we do not know $f(-1)$ - ? but f is a function from $[0, \infty)$ to $\mathbb{R}$

Let f and g be functions from A to $\mathbb{R}$.

Then $f + g$ and fg are also functions from A to $\mathbb{R}$ defined for all $x \in A$ by

$$(f + g)(x) := f(x) + g(x),$$

$$(fg)(x) = f(x)g(x).$$

For example let f_1 and f_2 be functions on $\mathbb{R}$ (from $\mathbb{R}$ to $\mathbb{R}$) such that

$$f_1(x) = x^2 \text{ and } f_2(x) = x - x^2.$$

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + x - x^2 = x$$

$$(f_1 f_2)(x) = f_1(x) f_2(x) = x^2(x - x^2) = x^3 - x^4$$

Question: Define the function $f_1 + f_2$, where $f_1(x) = \sqrt{x}$ and $f_2(x) = \frac{1}{x}$.

f_1 is defined on $[0, \infty)$, f_2 is defined on $\mathbb{R} - \{0\}$
 $f_1 + f_2$ can be defined on the intersection $(0, \infty)$

$$(f_1 + f_2)(x) = \sqrt{x} + \frac{1}{x}$$

$$\begin{aligned} -x &\geq 0 \\ \text{or } x &\leq 0 \end{aligned} \qquad \begin{aligned} x &\geq 1 \\ x &\geq 1 \end{aligned}$$

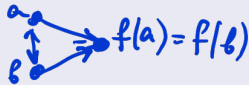
Question: Is it possible to define $f_1 f_2$, where $f_1(x) = \sqrt{-x}$ and $f_2(x) = \sqrt{x-1}$?

no

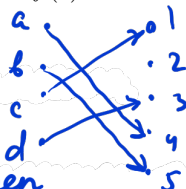


A function f is said to be **one-to-one** if and only if
 $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

A function is said to be **injective** if it is one-to-one.
 $\forall a \forall b (f(a) = f(b) \rightarrow a = b) \equiv 1$



The function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$ and $f(d) = 3$ is **one-to-one**



$f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is **not one-to-one**
 $f(1) = 1$ and $f(-1) = 1$



$f(x) = x + 1$ of $\mathbb{R}$ is **one-to-one**. If $f(a) = f(b)$, then
 $a + 1 = b + 1$ or $a = b$. Hence **one-to-one**.

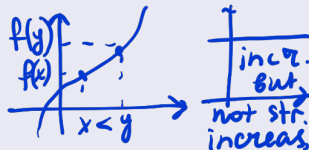
A function f is

increasing if $\forall x \forall y (x < y \rightarrow f(x) \leq f(y))$

strictly increasing if $\forall x \forall y (x < y \rightarrow f(x) < f(y))$

decreasing if $\forall x \forall y (x < y \rightarrow f(x) \geq f(y))$

strictly decreasing if $\forall x \forall y (x < y \rightarrow f(x) > f(y))$

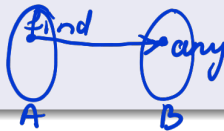


Strictly decreasing and strictly increasing functions are one-to-one. Answer Q3

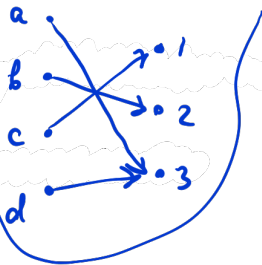
A function f from A to B is called **onto**, if and only if
for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

A function f is called **surjective** if it is onto.

$$\forall y \exists x (f(x) = y)$$



The function f from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ with $f(a) = 3$, $f(b) = 2$, $f(c) = 1$ and $f(d) = 3$ is onto.



$f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is **not onto**
 $\forall n \in \mathbb{Z}$ we can find $m \in \mathbb{Z}$ s.t. $f(m) = n$ or $m^2 = n$
 does not work for example for $n = -1$ or $n = 2$

$f(x) = x + 1$ of $\mathbb{R}$ is onto $\forall y \in \mathbb{R}$ there is $x \in \mathbb{R}$ s.t.

Answer Q4

$$f(x) = y$$

$$x + 1 = y$$

$$\boxed{x = y - 1}$$

$\Rightarrow$ onto

Summary

Suppose that $f : A \rightarrow B$.

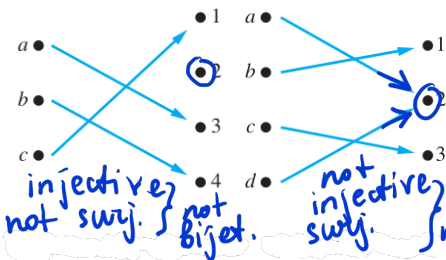
To show that f is **injective** show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not **injective** find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

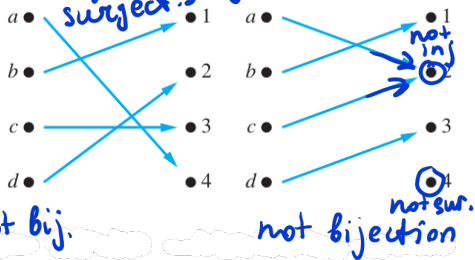
To show that f is **surjective** consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not **surjective** find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

A function that is injective and surjective at the same time is called **bijection** function, or **bijection**.



injective } bij.
surjective } bij.



Inverse functions



Let f be a bijection from the set A to the set B .

The inverse function of f is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$.

The inverse function of f is denoted by f^{-1} (not the same as $\frac{1}{f}$)

Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Bijection f is called **invertible**, since it has an inverse function.

f from $\{a, b, c\}$ to $\{1, 2, 3\}$ with $f(a) = 3$, $f(b) = 2$, $f(c) = 1$ is

$f(x) = x + 1$ of $\mathbb{R}$ is invertible

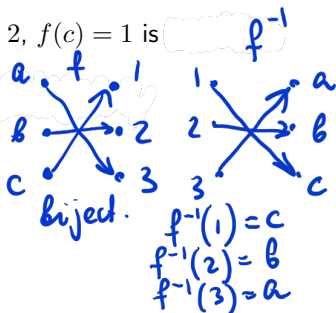
$$f^{-1}(x) = x - 1$$

$f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is not biject.

but $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is

not onto since $1 \rightarrow 2$
not bijection
not invert.

Answer Q5



Composition



Let $g : A \rightarrow B$ and $f : B \rightarrow C$.

The **composition** of f and g is the function $f \circ g : A \rightarrow C$ defined by

$$(f \circ g)(a) = f(g(a))$$

Let $f(x) = 2x + 3$ and $g(x) = x^2$ on $\mathbb{R}$.

$$(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 + 3$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = (2x + 3)^2 = 4x^2 + 12x + 9$$

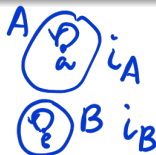
$$f \circ g \neq g \circ f.$$

Let $f : A \rightarrow B$ be invertible.

For $a \in A$ with $f(a) = b$ we have

$$(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a$$

$$(f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b$$



$$f^{-1} \circ f = i_A \quad f \circ f^{-1} = i_B$$

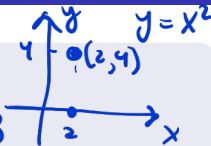
Answer Q6

The graph of functions

Let $f : A \rightarrow B$.

The **graph** of f is the set

$$\{(a, b) \mid a \in A, f(a) = b\} \subset A \times B$$



The graph of the function $f(n) = 2n + 1$ on the set of integers is at Figure 1

The graph of the function $f(n) = n^2$ on the set of integers is at Figure 2

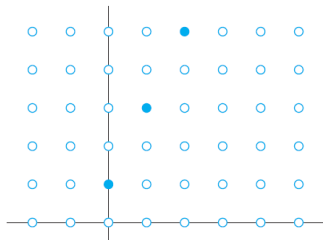


Figure 1

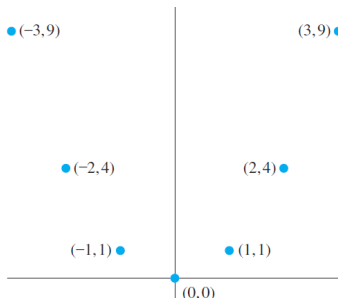


Figure 2

Floor and ceiling functions

The **floor** function assigns to the real number x the largest integer that is less than or equal to x .

The floor function is denoted by $\lfloor x \rfloor$

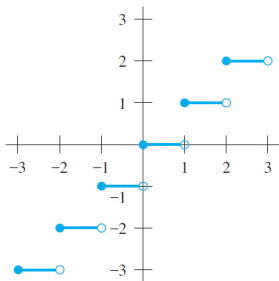
The **ceiling** function assigns to the real number x the smallest integer that is less than or equal to x .

The ceiling function is denoted by $\lceil x \rceil$

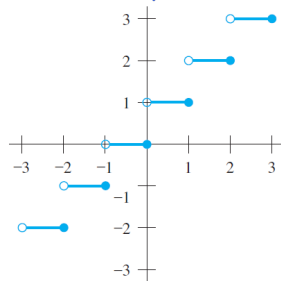
$$\lfloor 5.2 \rfloor = 5 \quad \lfloor -3.1 \rfloor = -4$$



$$\lceil 5.2 \rceil = 6 \quad \lceil -3.1 \rceil = -3$$



$\lfloor x \rfloor$



$\lceil x \rceil$

Answer Q7

Homework:

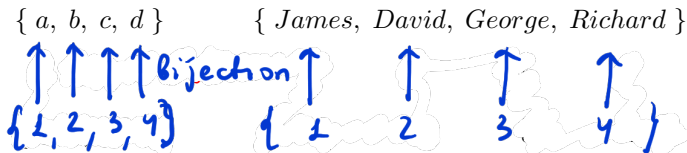
- Solve all *odd* exercises starting at p. 152.

Cardinality of sets

(paragraph 2.5)

The same cardinalities

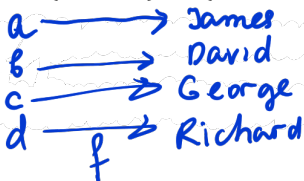
Question: When two **finite** sets A and B have the same cardinality?



Two (finite or infinite) sets A and B are said to have **the same cardinality** if there is a bijection between them. In this case we write $|A| = |B|$.

Construct a bijection

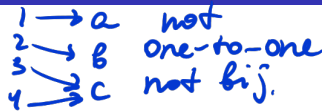
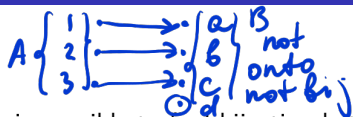
$$f : \{a, b, c, d\} \rightarrow \{James, David, George, Richard\}$$



Can construct many bijections here.
to compare cardin. constr. just one bij.

Answer Q8

Different cardinalities



It is impossible to find bijection between sets with different cardinalities. !!!

In this case we write $|A| \leq |B|$. There is a one-to-one function from A to B .

If there is a one-to-one function from A to B , we write $|A| \leq |B|$.

When $|A| \leq |B|$ and A and B have different cardinality, we write $|A| < |B|$.

However, it is not easy to show that two infinite sets have different cardinality.

For example, we can construct a one-to-one function from $\mathbb{N}$ to $\mathbb{Z}$, hence

$|\mathbb{N}| \leq |\mathbb{Z}|$. Similarly, $|\mathbb{N}| \leq |\mathbb{R}|$.

We will show that $|\mathbb{N}| = |\mathbb{Z}|$ but $|\mathbb{N}| < |\mathbb{R}|$

$$\mathbb{N} = \{0, 1, 2, 3, \dots\} \quad \mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$$

Answer Q9

$$f(n) = n, \quad n \in \mathbb{N}$$

$f: \mathbb{N} \rightarrow \mathbb{Z}$ one-to-one

1

Countable sets

$\mathbb{Z}^+$

not $\mathbb{Z}^+$

We will now split all infinite sets into two groups, those with the same cardinality as the set of positive integers and those with a different cardinality.

A set that is either finite or has the same cardinality as the set of positive integers is called countable. $\equiv |\mathbb{Z}^+|$ ($\mathbb{Z}^+ = \{1, 2, 3, \dots\}$)

A set that is not countable is called uncountable.

When an infinite set S is countable, we write $|S| = \aleph_0$ (where $\aleph$ is aleph, the first letter of the Hebrew alphabet).

Question: Show that the set of odd positive integers is a countable set.

Solution:

$f(n) = 2n - 1$ $f: \mathbb{Z}^+ \rightarrow \mathbb{O}^+$ compare $\mathbb{Z}^+$ and $\mathbb{O}^+$

1. Let $f(n) = f(m)$. Then $2n - 1 = 2m - 1$
or $2n = 2m$ or $n = m \Rightarrow$ one-to-one

2. Let $t \in \mathbb{O}^+$, then $t = 2n - 1$ for some $n \in \mathbb{Z}^+$

Hence, $t = f(n) \Rightarrow$ onto

$\Rightarrow f$ bijection $\Rightarrow |\mathbb{O}^+| = |\mathbb{Z}^+|$
 $\Rightarrow \mathbb{O}^+$ is count.

1	$\rightarrow$	1
2	$\rightarrow$	3
3	$\rightarrow$	5
$\vdots$		$\vdots$

HILBERT'S GRAND HOTEL



The famous mathematician David Hilbert invented the notion of the Grand Hotel, which has a **countably infinite number of rooms**, each occupied by a guest.

When a new guest arrives at a hotel with a finite number of rooms, and all rooms are occupied, this guest cannot be accommodated without evicting a current guest. However, we can always accommodate a new guest at the Grand Hotel, even when all rooms are already occupied

Because the rooms of the Grand Hotel are countable, we can list them as **Room 1, Room 2, Room 3, and so on**. When a new guest arrives, we move the guest in Room 1 to Room 2, the guest in Room 2 to Room 3, and in general, the guest in Room n to Room $n + 1$, for all positive integers n . This frees up Room 1, which we assign to the new guest, and all the current guests still have rooms.

Examples of countable sets

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

$$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- The set of positive integers $\mathbb{Z}^+$, with bijection

- The set of natural numbers $\mathbb{N}$, with bijection

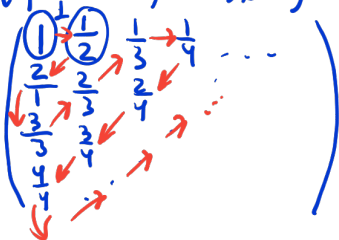
- The set of all integers $\mathbb{Z}$, with bijection

$$f(n) = \frac{n-1}{2} \text{ if } n \text{ is odd}$$

- The set of positive rational numbers, with bijection

$$= \left\{ \frac{m}{n} \mid m \in \mathbb{N}, n \in \mathbb{Z}^+ \right\}$$

Answer Q10

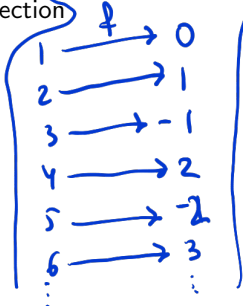


$$f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+ \quad f(n) = n$$

$$f: \mathbb{N} \rightarrow \mathbb{Z}^+ \quad f(n) = n+1$$

$$f(n) = \frac{n}{2} \text{ if } n \text{ is even}$$

$$f: \mathbb{Z}^+ \rightarrow \mathbb{Z}$$



Countable union of countable sets

Theorem

If the sets $S_1, S_2, S_3, \dots$ are countable, then the countable union

$$\bigcup_{i=1}^{\infty} S_i$$

}!

is countable.

Example from midterm

(8 points) Let M_n be the set of $n \times n$ matrices with entries that are either 0 or 1.

(a) (2 point) Give two examples of matrices from the set M_2 . $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

(b) (2 points) Is the set M_2 finite or infinite? Justify.

(c) (4 points) Let M be a set of all square matrices (of all sizes) with entries, that are either zero or one. Is the set M finite, countably infinite or uncountable? Justify.

$M = \bigcup_{n=1}^{\infty} M_n$ The countable union of finite sets is countable.

Uncountable sets

Theorem

The interval $(0, 1)$ is an uncountable set

Proof: (Cantor diagonalization argument)

Assume that $(0, 1)$ is countable. It means that we can write them all as

$$r_1 = 0.\overset{\circ}{d}_{11}d_{12}d_{13}d_{14}\dots \quad 0.5831\dots \in (0, 1)$$

$$r_2 = 0.d_{21}\overset{\circ}{d}_{22}d_{23}d_{24}\dots$$

$$r_3 = 0.d_{31}d_{32}\overset{\circ}{d}_{33}d_{34}\dots$$

$$r_4 = 0.d_{41}d_{42}d_{43}d_{44}\dots$$

$\vdots$

where $d_{ij} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Let $r = 0.d_1d_2d_3d_4\dots$, where

$$\begin{matrix} r_1 & \neq & r \\ r & \neq & r_2 \end{matrix} \dots$$

$$d_1 = \begin{cases} 4, & d_{11} \neq 4 \\ 5 & d_{11} = 4 \end{cases}$$

r is in $(0, 1)$ but $r \neq r_i$ for all i . Hence, $(0, 1)$ is not countable.

Examples of uncountable sets

Show that $|(0, \infty)| = |(0, 1)|$

Consider the function $f : (0, \infty) \rightarrow (0, 1)$ given by

$$f(x) = \frac{x}{x+1}.$$

Show that $|\mathbb{R}| = |(0, 1)|$

Show that $|\mathbb{R}| = |(0, \infty)|$.

Consider the function $f : \mathbb{R} \rightarrow (0, \infty)$ given by

$$f(x) = 2^x.$$

SCHRÖDER-BERNSTEIN THEOREM

If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

In other words, if there are one-to-one functions f from A to B and g from B to A , then $|A| = |B|$.

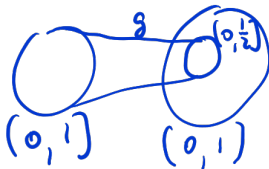
Show that $|(0, 1]| = |(0, 1)|$

$f(x) = x$ is a one-to-one function from $(0, 1)$ to $(0, 1]$.

The function $g(x) = x/2$ is one-to-one from $(0, 1]$ to $(0, 1/2] \subset (0, 1)$.

By the S-B theorem, $|(0, 1)| = |(0, 1]|$.

Answer Q11



The Set of all Subsets of Natural Numbers is Uncountable

$$|\mathcal{P}(A)| = 2^{|A|}$$

Theorem 1: The set $\mathcal{P}(\mathbb{N})$ of all subsets of $\mathbb{N}$ is uncountable.

$$|A| = n$$

In the proof below, we use the famous diagonalization argument to show that the set of all subsets of $\mathbb{N}$ is uncountable.

- **Proof:** Suppose that $\mathcal{P}(\mathbb{N})$ is countable.
- We can denote each set $A \subseteq \mathbb{N}$ by a decimal number of the form:

$$0.x_0x_1 \dots x_n \dots$$

- where each $x_n \in \{0, 1\}$ and such that $x_n = 0$ if $n \in A$ and $x_n = 1$ if $n \notin A$. For example, the set $\{1, 3, 4, 5, 6\}$ has decimal representation 0.1011110.... Now since we assume that $\mathcal{P}(\mathbb{N})$ is countable, there exists a bijective function $f: \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ defined for all $n \in \mathbb{N}$ by $f(n) = A_n$. We list the sets $\{A_1, A_2, \dots, A_n, \dots\}$ and their decimal representations as follows:

$$A_1, 0.x_{1,1}x_{1,2} \dots x_{1,n} \dots$$

$$A_2, 0.x_{2,1}x_{2,2} \dots x_{2,n} \dots$$

$$\vdots$$

$$A_n, 0.x_{n,1}x_{n,2} \dots x_{n,n} \dots$$

$$\vdots$$

- Now construct a decimal number $x = 0.x_1x_2 \dots x_n \dots$ as follows. We let $x_1 = 0$ if $x_{1,1} = 1$ and we let $x_1 = 1$ if $x_{1,1} = 0$. We let $x_2 = 0$ if $x_{2,2} = 1$ and we let $x_2 = 1$ if $x_{2,2} = 0$. In general, we let $x_n = 0$ if $x_{n,n} = 1$ and we let $x_n = 1$ if $x_{n,n} = 0$. Then X differs from all of the decimal numbers in the list above in at least one decimal spot. So the decimal x corresponds to a subset X of $\mathbb{N}$ that is not on the list above. This is a contradiction since f is assumed to be a bijection.
- Therefore the set $\mathcal{P}(\mathbb{N})$ is uncountable. ■

If $A = \mathbb{N}$
what
is $2^{|\mathbb{N}|}$ (1)

cardinality
of uncount.
set (2)

$\Rightarrow |\mathcal{P}(\mathbb{N})|$ - uncount.

Uncomputable functions



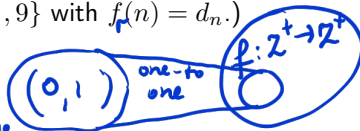
We say that a function is **computable** if there is a computer program in some programming language that finds the values of this function.

If a function is not computable we say it is **uncomputable**.

To show that there are uncomputable functions, we need to establish two results:

- 1 The set of all computer programs in any particular programming language is countable (a computer programs is a string of characters from a finite alphabet)
- 2 the set of functions from the set of positive integers to itself is uncountable (construct a one-to-one correspondence between the set $(0, 1)$ and a subset of these functions. Do this by associating to the real number $0.d_1d_2 \dots d_n \dots$ the function $f_r: \mathbb{Z}^+ \rightarrow \{0, 1, \dots, 9\}$ with $f_r(n) = d_n$.)

If $f_{r_1} = f_{r_2} \Rightarrow$
 d_n of r_1 are d_n of r_2
 $\Rightarrow r_1 = r_2 \Rightarrow$ one-to-one



The continuum hypothesis

We know that $\mathbb{N}$ is countable that is

$$|\mathbb{N}| = \aleph_0$$

The power set $\mathcal{P}(\mathbb{N})$ is uncountable, that is

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} = \underline{\underline{c}}$$

The continuum hypothesis

There is **no** set A such that

$$\aleph_0 < |A| < c$$

The continuum hypothesis was advanced by Georg Cantor in 1878, and published as the first of Hilbert's 23 problems presented in 1900.

Homework:

- paragraphs 2.4 (sequences) and 2.6 (matrices) are left for independent study
- Solve *odd* exercises starting at p. 176, #1-#25 and #33, #35.